F005 | External wave-data benchmark

v0.4.2 qualification protocol - 20 October 2026 - DEVELOPMENT / NON-BLIND

Purpose: qualify conditional multistep predictive performance and audit waveform-family overlap, while preserving the existing candidate split, prior releases, source archives, and TEST/OOD elevation secrecy.

1. Locked factual starting point: v0.4.1

Measure	v0.4.1 recorded result
Validation trajectories	750
Anchors	0.004, 1.5, 2.8 s
Horizons	0.1, 0.5, 1.0, 2.0, 3.0 s
ARX, 1.5 s anchor, 3 s endpoint	0.778 mm RMSE
Persistence, same endpoint	3.625 mm RMSE
Generated input-shape pairs / scored	75,181 / 15,000
Near-shape cross-partition pairs in scored subset	4,800
Certification	NOT BLIND; PKLSA/Euler NOT RUN

Scope warning: the v0.4.1 RMSE figures are errors at selected forecast endpoints, not the average across the entire predicted path. The earlier endpoint win does not establish a full-path win against a zero-equilibrium baseline.

2. Exact v0.4.2 implementation commitments

Gate	Qualification action / limit
P5c	Score all generated input-fingerprint candidate pairs with same normalized cross-correlation criteria, without 15,000 cap. Not an exhaustive all-pairs or source-family audit.
P6d	Report endpoint and full-path RMSE at all anchors/horizons; no water-height teacher forcing within horizon.
P6e	Benchmark ARX against persistence, linear extrapolation, zero equilibrium and unforced damped oscillator. Oscillator parameters derive from TRAIN-only ARX.
P6f	Descriptive phase/amplitude at locked ARX frequency, with minimum-amplitude threshold.
P6g	Report candidate-linked VAL-TRAIN overlap and screen-negative validation sensitivity; original split unchanged.

Reproducibility, controls and next gate

3. Dimensional contract and prediction recurrence

The experiment reports acceleration in m/s^2 and elevation in mm. Prediction calculations convert elevation to m. With dt = 0.004 s:

```
eta[n+1] = A*eta[n] + B*eta[n-1] + C*dt^2*a[n]
```

Locked TRAIN-only model: A=1.985666104, B=-0.990869642, C=-1.118156918. Two measured heights are used only at each predeclared origin; the future acceleration remains known as exogenous input.

Path error: RMSE computed over every trajectory and every intermediate timestep up to the horizon. **Endpoint error:** RMSE computed only at the horizon endpoint. Both are reported in mm.

4. Separation of data and interpretation

Allowed	Forbidden
Read all acceleration trajectories (input-only screening).	Read TEST/OOD water-height trajectories or compute their metrics.
Read VALIDATION heights for fixed predeclared metrics.	Retrain or tune the frozen ARX on validation labels.
Score all proposed input-shape pairs.	Claim complete all-pairs coverage or publisher-family independence.
Fit nothing new; use locked TRAIN-derived oscillator controls.	Promote an oscillator/control to independent Euler or PKLSA validation.

5. Expected local outputs

Within `outputs/f005_qualification_v0.4.2/`: JSON provenance report, summary Markdown, horizon-metrics CSV, matched-family edges, full input-shape component assignment CSV, input-screen-negative validation sensitivity CSV, phase/amplitude CSV, sample trajectories, and a generated PDF.

6. Qualification decision

P5c: REVIEW until source-defined family IDs or independently justified complete family mapping is available. **P6d/e:** descriptive/conditional development validation only, even if path RMSE improves. **P6g:** group-disjoint certification BLOCKED; TEST/OOD remain sealed. **P7:** PKLSA NOT RUN. **P8:** SST physical claims UNRESOLVED.

References: Giulietti et al. (2026), Robot-executed sloshing dataset, Zenodo DOI: 10.5281/zenodo.20039757. Ljung (1999), System Identification: Theory for the User, 2nd ed., Prentice Hall, ISBN 9780136566953.